NATIONAL UNIVERSITY OF SINGAPORE

EE2213 - Introduction to Artificial Intelligence

Semester 1: AY 2025/26

Quiz 4: L12-L13

Time Allowed: 30 minutes

INSTRUCTIONS TO STUDENTS

1. This assessment consists of **TWO** types of questions: Multiple Choice Questions (MCQs, only one correct answer), and Multiple Response Questions (MRQs, allow multiple correct answers).

2. There are a total of **10 Questions** consisting of **4** Multiple Choice Questions (*3 marks each*), and **6** Multiple Response Questions (*5 marks each*) for a total of **42** marks.

3. Please answer **ALL** questions.

4. This is a **NON-SECURE BLOCK INTERNET** assessment. You may use printed or soft copy materials (e.g., lecture slides, textbooks, notes). However, you are not allowed to search the internet or use any LLM-powered tools (e.g., ChatGPT or other language models)

5. Electronic devices other than your PC should be left on the floor next to your desk or put inside your bags at all times. Please note that graphing calculators are not permitted. A basic calculator is provided for numerical calculations, accessible under "Tool Kit" at the top right of the exam interface

6. You are not permitted to leave the venue for the first 15 minutes.

7. If you wish to communicate with an invigilator, go to the washroom, or leave before the end of the assessment, please raise your hand to inform the invigilator.

8. You will be liable for disciplinary action which may result in expulsion from the University if you are found to have contravened any of the clauses below,

- Violation of the NUS Code of Student Conduct (in particular the part on Academic, Professional and Personal Integrity), NUS IT Acceptable Use Policy or NUS Examination rules.
- Possession of unauthorized materials/electronic devices.

- Bringing your mobile phone or any storage/communication device with you to the washroom.
- Unauthorized communication e.g. with another student.
- Use locally installed Large Language Models (LLMs) during the assessment.
- Reproduction of any exam materials outside of the exam venue.
- Photography or videography within the exam venue.
- Plagiarism, giving or receiving unauthorised assistance in academic work, or other forms of academic dishonesty.

9. Once you have completed the assessment,

- Click on the "Exam Controls" button and choose "Submit Exam".
- Check off “I am ready to exit my exam” and click on "Exit" to upload your answers to the server.
- You will see a green confirmation window on your screen when the upload is successful. **Please keep this window on your screen.**
- If you do not see a green window, please disconnect and reconnect your WIFI and try again.
- Please be reminded that it is your responsibility to ensure that you have uploaded your answers to ExamSoft.
- Please remain seated quietly in the meantime.

10. By clicking on the "Continue" button below, I acknowledge that I have read and fully understood this Exam Notice.

Thank You

Question 1

(MCQ) The rules of chess (e.g., the bishop moves diagonally”) are an example of:

(a) Declarative knowledge
(b) Procedural knowledge
(c) Heuristic knowledge
(d) Meta-knowledge

Answer: (a). These are explicit facts/rules.

Question 2

(MCQ) Which type of knowledge representation struggles the most with uncertainty?

(a) Logical
(b) Structured
(c) Probabilistic
(d) Distributed

Answer: (a). Logic requires absolute truth values; it does not natively support uncertainty.

Question 3

(MCQ) When Sherlock Holmes says, *"The dog did not bark, therefore the intruder must have been someone it knew"*, he is using:

(a) Deductive reasoning
(b) Inductive reasoning
(c) Abductive reasoning
(d) None

Answer: (c). "Intruder must have been someone it knew" is the most plausible explanation of "The dog did not bark".

Question 4

(MRQ) Which of the following is/are a well-formed formula (WFF) in propositional logic?

(a) $\alpha \equiv \beta$, provided that α and β are both well-formed formulas.
(b) $\neg\neg\neg\sigma \Leftrightarrow \neg\beta \vee \alpha$, provided that α, β and σ are all well-formed formulas.
(c) $(A \vee B) \ \wedge \neg C$, provided that A, B and C are proposition symbols.
(d) $\neg A \Leftrightarrow B \vee C \wedge D \Rightarrow\Rightarrow E$, provided that A, B, C, D and E are all proposition symbols.
(e) α

Answer: (b)(c)

(a) is wrong because $\equiv$ is not part of syntax of propositional logic. It belongs to semantics.

(d) is wrong because $\Rightarrow\Rightarrow$ is syntactically wrong in propositional logic.

(e) is wrong because we are not sure if α is a WFF.

Question 5

(MRQ) Which of the following is/are a valid knowledge base in propositional logic? Assume P, Q, R and S are proposition symbols.

(a) {¬P, Q, $Q \Leftrightarrow P$}
(b) {¬¬P, P∨ $Q \wedge R, \neg R$}
(c) {P∨ $\neg Q \Leftrightarrow \neg, Q \wedge R$}
(d) {P⇒ P ∨Q, P∧ $R \vee Q \wedge S$, Q⇔ S}
(e) None

Answer: (b)(d). Check if KB is satisfiable and each formula inside is a well-formed formula.

Question 6

(MCQ) How many models satisfy P∧ $Q \Leftrightarrow \neg Q \vee R$ for 3 proposition symbols P, Q and R.

(a) 0
(b) 1
(c) 2
(d) 3
(e) 4

Answer: (c)

Model 1= {P=True, Q=True, R=True}

Model 2 = {P=False, Q=True, R=False}

Question 7

(MRQ) Three friends, Alice, Bob, and Carol, are deciding who will attend a party. Consider the following statements:

1. If Alice goes to the party, then Bob will also go.
2. If Carol goes to the party, Alice will not go.
3. Either Bob or Carol goes to the party (or both).

Let A= "Alice goes", B= "Bob goes", C= "Carol goes". Which of the following statements is/are correct?

(a) There are 4 models that satisfy all the 3 statements simultaneously for 3 proposition symbols A, B and C.

(b) If all the 3 statements are true, it is possible that Alice goes to the party.
(c) If all the 3 statements are true, Bob must go to the party.
(d) If all the 3 statements are true, it is possible that Carol goes to the party.
(e) None

Answer: (a)(b)(d)

Please see "Quiz 4_A.ipynb"

Question 8

(MRQ) Given the following:

"If the unicorn is mythical, then it is immortal, but if it is not mythical, then it is a mortal mammal. If the unicorn is either immortal or a mammal (not both), then it is horned. The unicorn is mythical if it is horned."

Which of the following statements is/are correct?

(a) The unicorn is mythical.
(b) The unicorn is immortal.
(c) The unicorn is horned.
(d) The unicorn is a mammal.
(e) None

Answer: (a)(b)

Please see "Quiz 4_A.ipynb"

Question 9

(MRQ) Which of the following is/are correct? Assume A, B and C are proposition symbols.

(a) $(A \land B) \models (A \Rightarrow B)$
(b) $(A \Leftrightarrow B) \models (A \lor B)$
(c) $(C \lor (\neg A \land \neg B)) \equiv ((A \Rightarrow C) \land (B \Rightarrow C))$
(d) $\frac{\neg(A \lor (B \land \neg C))}{\neg A \land \neg B \lor \neg A \land C}$
(e) The number of the models satisfying $(A \Leftrightarrow B) \Leftrightarrow C$ is the same as the number of the models satisfying $(A \Leftrightarrow B)$ for any fixed set of proposition symbols that includes A, B and C.

Answer: (a)(c)(d)(e)

(a) and (b) can look at the truth tables and check if the right-hand side is true in every model where left-hand side is true.

(c): $(C \vee(\neg A \wedge \neg B)) \equiv (C \vee \neg A) \wedge (C \vee \neg B) \equiv (A \Rightarrow C) \wedge (B \Rightarrow C)$.

(d): $\neg(A \vee (B \wedge \neg C)) \equiv \neg A \wedge \neg(B \wedge \neg C) \equiv \neg A \wedge (\neg B \vee \neg C) \equiv \neg A \wedge \neg B \vee \neg A \wedge C$. The bottom part and top part are logically equivalent

(e): Please see "Quiz 4_A.ipynb"

Question 10

(MRQ) Three friends: Alice, Bob, and Carol. Each either always tells the truth or always lies (not both).

Alice says: "If Bob tells the truth, then Carol lies."

Bob says: "Either Alice or Carol tells the truth (not both)."

Carol says: "Alice lies."

Which of the following is/are correct?

(a) Alice lies.
(b) Bob tells the truth.
(c) Carol lies.
(d) None

Answer: (b)

Please see "Quiz 4_A.ipynb"